

Section header 1

CS-433 Machine Learning: Project 1

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I. INTRODUCTION

This report will discuss the data preprocessing, accuracy metrics, and machine learning methods applied to the Behavioral Risk Factor Surveillance System (BRFSS). In particular, we implement a neural network to accurately predict new cases. Our goal is to predict whether an individual is likely to develop cardio-vascular disease (CVD) based on clinical and lifestyle factors.

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II. PRELIMINARY MODELS

Six basic models were implemented: gradient descent, stochastic gradient descent, least squares regression using normal equations, ridge regression, logistic regression, logistic regression, and regularized logistic regression. While the accuracy of predictions appeared high, the corresponding F1-score was low. This motivates us to gain a better understanding of the data and underlying characteristics.

III. DATA PROFILE AND CLEANING

The BRFSS dataset includes a wide range of demographic, behavioral, and medical variables. Each record represents one survey respondent, while each column corresponds to a specific attribute. The target variable MICHD is binary:

$$\text{MICHD} = \begin{cases} 1, & \text{if the respondent has been diagnosed with coronary heart disease,} \\ -1, & \text{otherwise.} \end{cases}$$

This setup defines a binary classification problem. A major challenge of this dataset is the strong imbalance between positive and negative labels. As shown in Figure 1a, respondents are overwhelming classified as negative (no MICHD). Another challenge lies in missing or incomplete values. Several features contain null or undefined values, reducing the reliability of the dataset, as shown in Figure 1b. Lastly, the dataset contains variables with different numerical scales and measurement units. For example, age is measured in years, BMI is continuous, and categorical survey responses are encoded numerically. Such scale differences can distort model training, particularly for distance-based or gradient-based algorithms.

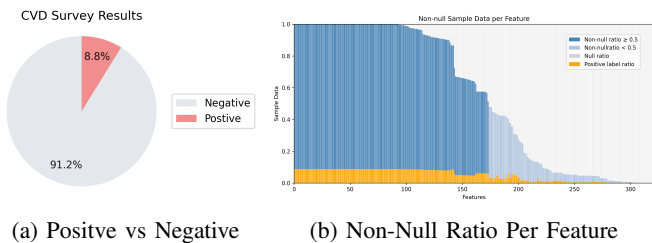


Figure 1: Data Visualization

Based on these challenges, we take the following steps to clean and standardize the data. First, the ID columns are removed as they do not contain predictive information. Missing values are replaced with zero to ensure numerical consistency. Features with null value ratios over 0.5 are dropped due to the irregularity of positive labels. All features were standardized using z-score normalization to stabilize training. We convert the labels into binary form, assigning 0 to non-positive values and 1 to positive values. These steps show an improvement in data quality and stability in model training.

IV. NEURAL NETWORK

Linear regression often fails to achieve satisfactory performance on nonlinear data. It is also not extendable, as any ensemble of simple linear models remains fundamentally linear (a consequence of simple algebra). To capture complex relationships and introduce nonlinearity into the model, we drop

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